23-S2-Q2

Q: class 1 & -1

(a) $x(N)$

$d(N)$

SVM formulate the primel problem

separable

(2) SVM → x.

$S_1 = [0.2575 \quad 0.0598 \quad 0.7212 \quad 0.8555]^T$

$S_2 = [4.5052 \quad -0.4733 \quad 3.3539 \quad 4.6830]^T$

$S_3 = [3.5651 \quad -1.3827 \quad 2.8324 \quad 2.6521]^T$

$S_4 = [-0.2015 \quad -2.1464 \quad 1.0908 \quad 1.5509]^T$

Solution (a) Given the training sample $\{x(i), d(i)\}_{i=1,2,\cdots,N}$, find the optimal values of the weight vector w and bias b such that they satisfy the constraints

$$d(i) \times [w^T x(i) + b] \geq 1 \quad \text{for } i = 1, 2, \cdots, N$$

and the weight vector minimizes the following cost function

$$J(w) = \frac{1}{2} w^T w$$

(b) ① calculate w^*

$$w^* = \sum_{i=1} \alpha(i) d(i) s(i)$$

$$= 0.3646 \times 1 \times [0.2575 \quad 0.0598 \quad 0.7212 \quad 0.8555]^T$$
$$+ 0.0067 \times (-1) \times [4.5052 \quad -0.4733 \quad 3.3539 \quad 4.6830]^T$$
$$+ 0.029 \times (-1) \times [3.5651 \quad -1.3827 \quad 2.8324 \quad 2.6521]^T$$
$$+ 0.3289 \times (-1) \times [-0.2015 \quad -2.1464 \quad 1.0908 \quad 1.5509]^T$$

$$= [-9.2042 \quad 4.3511 \quad -7.5341 \quad -7.1733]^T$$

② calculate b^*

$$b^* = \frac{1}{N_s} \sum_{\alpha(i) \neq 0} [d(i) - (w^*)^T x(i)]$$

$$= \frac{1}{4} \times [1 - (w^*)^T x(1) - 1 - (w^*)^T x(2) - 1 - (w^*)^T x(3) - 1 - (w^*)^T x(4)]$$

$$= \frac{1}{4}\left\{-2 - (w^*)^T\left[x(1) + x(2) + x(3) + x(4)\right]\right\}$$

$$= \frac{1}{4}\left\{-2 - \begin{bmatrix}-9.2042 & 4.3511 & -7.5341 & -7.1733\end{bmatrix}\begin{bmatrix}8.1263 \\ -3.9426 \\ 7.9983 \\ 9.7415\end{bmatrix}\right\}$$

$$= 55.0224$$

③ $y = w^T x + w_0$

$$= (w^*)^T x + b^*$$

$$= \begin{bmatrix}-9.2042 & 4.3511 & -7.5341 & -7.1733\end{bmatrix}\begin{bmatrix}-0.2083 \\ -2.2553 \\ 2.6182 \\ 2.6170\end{bmatrix} + 55.0224$$

$$= 8.5823 > 0$$

predicted class = 1

(c) Fisher's Linear Discriminant Analysis (LDA)

Linear SVM

① criterion

LDA: find such w that the following criterion is maximized

$$J(w) = \frac{(\tilde{m}_1 - \tilde{m}_2)^2}{\tilde{S}_1^2 + \tilde{S}_2^2}$$

SVM: find the weight vector minimizes the following cost function $J(w) = \frac{1}{2} w^T w$

② philosophy

LDA : max the ratio $\frac{w^T S_B w}{w^T S_w w}$

SVM : max the margin between the closest point of opposite classes

③ Data Usage

LDA : all sample mean and covariances influence w

SVM : Only support vectors determine w

④ Optimisation

LDA : Closed-form eigen vector solution for two classes

SVM : convex quadratic programme solved via Lagrange

⑤ Summary

LDA : optimises class separation in a probabilistic sense und Gaussian assumption

SVM : optimises geometric margin with minimal distributional assumption.